

# GPU scaling runs: ALIGNN-FF and VASP

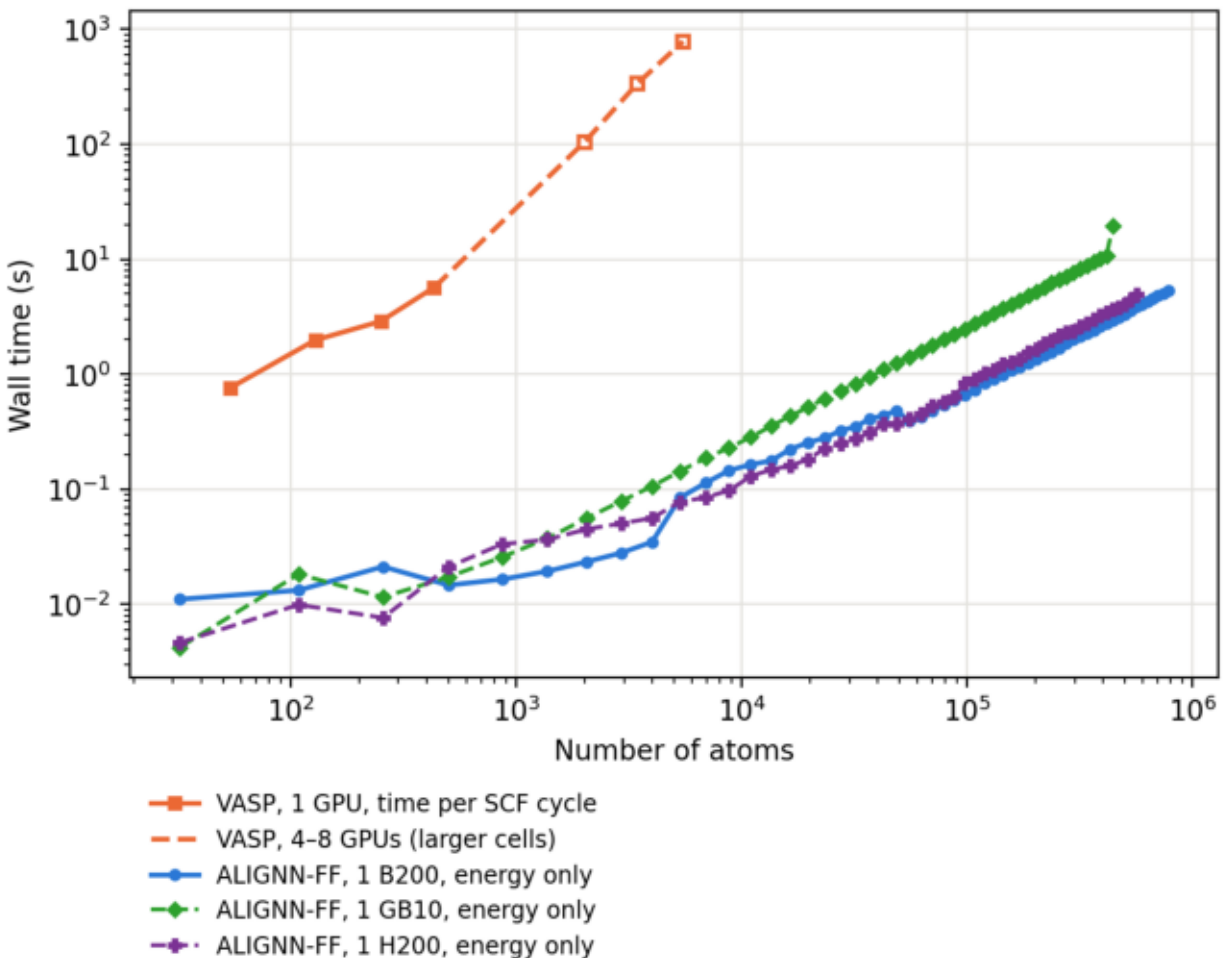
B200 runs: NVIDIA B200 GPUs (SLURM partition b200, QOS blackwell\_test).

ALIGNN-FF, energy only (no forces), Cu FCC, 1 B200: 58 supercells,  $1\times 1\times 1$  to  $58\times 58\times 58$  (4 to 780,448 atoms). Energies correct up to 442,368 atoms ( $48\times 48\times 48$ ); larger sizes are affected by the float32 drift and are shaded red in the table.

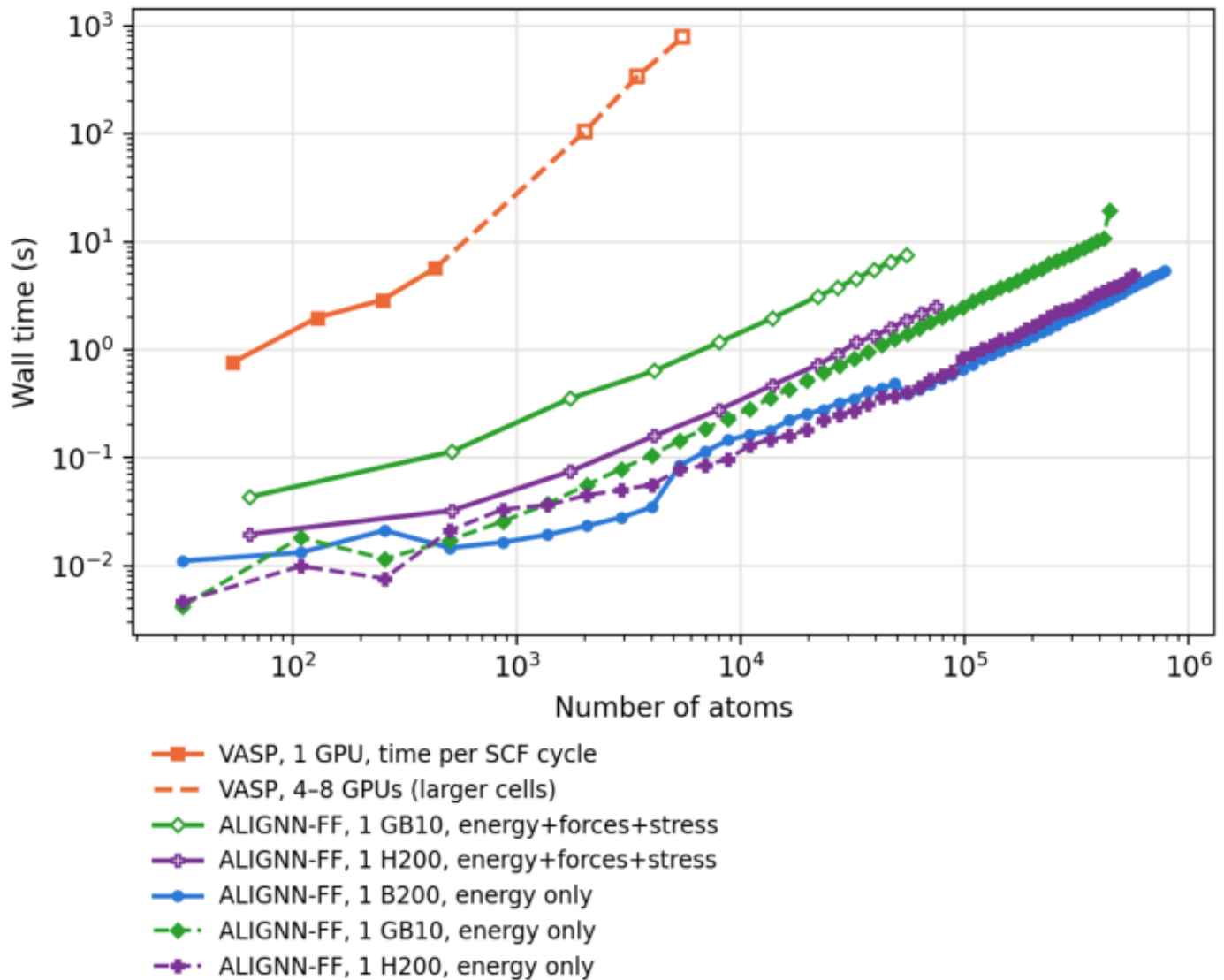
VASP, Si diamond: 21 runs over 9 supercell sizes (54 to 8,192 atoms); 19 completed, 2 crashed before the first SCF cycle ( $15\times 15\times 15$  and  $16\times 16\times 16$ ).

ALIGNN-FF, energy only, B200 settings rerun on a GB10 and an H200 (Cu FCC, same checkpoint, pure-PyTorch model; energies match the B200 to  $4e-6$  eV/atom); largest cell 442,368 atoms (GB10) and 562,432 (H200). At 256,000 atoms, inference takes 5.48 s (GB10), 0.70 s (H200) and 0.62 s (B200).

ALIGNN-FF with energy + forces + stress (added 2026-09-27; the MD workload), Si diamond, 1 GPU, default matpes\_r2scan checkpoint (hidden 128, smooth cutoff, 52 neighbours), same script on both cards: GB10 (121.7 GiB): largest cell 54,872 atoms at 107 GB, OOM at 64,000; H200 NVL (139.8 GiB): 74,088 atoms at 145 GB, OOM at 85,184. Memory 1.95 MB/atom on both; the H200 is 4× faster than the GB10 at every size.



## All runs: VASP and ALIGNN-FF on B200, GB10 and H200



Energy only: filled markers (B200 run and its GB10/H200 reruns, Cu, 12 neighbours). Energy + forces + stress: hollow markers, solid lines (Si, smooth 52-neighbour matpes\_r2scan, the MD workload). VASP: time per SCF cycle, Si.

## Which GPU for which job

|                                                  | GB10                | H200 NVL  | B200      |
|--------------------------------------------------|---------------------|-----------|-----------|
| Energy only, 256,000 atoms: inference (s)        | 5.48                | 0.70      | 0.62      |
| Energy only, 256,000 atoms: graph build (s, CPU) | 1.12                | 1.50      | 1.07      |
| Energy only: largest cell (atoms)                | 442,368             | 562,432   | 780,448   |
| Energy + forces + stress, 21,952 atoms (s)       | 3.12                | 0.72      | not run   |
| Energy + forces + stress: largest cell (atoms)   | 54,872              | 74,088    | not run   |
| GPU memory                                       | 121.7 GiB (unified) | 139.8 GiB | 178.3 GiB |

### Reading the numbers

- Model evaluation: the H200 is within 13% of the B200 and 8x faster than the GB10 (energy only, identical settings and checkpoint). With forces and stress the H200 is 4.3x the GB10.
- Graph construction runs on the host CPU, so it follows the machine, not the GPU.
- The largest cell is set by memory alone: 1.95 MB per atom with forces, 0.25 MB/atom energy only, the same on the GB10 and H200.

### Recommendation

- Long training runs: GB10. Slowest per GPU, but locally available at no cost; multi-node training, which would recover much of the per-GPU gap, is still being made to work.
- Large-cell MD and fast turnaround: H200. Largest cell with forces we can reach (74k atoms) and ~4x the GB10, but it is a shared, allocation-billed resource.
- B200: fastest and largest on paper, but not available to this project; the H200 comes within 13% of it on model evaluation, so it is not worth pursuing.

Caveats: single-point evaluations, not full MD or training steps, so the ratios will shift somewhat for real workloads; multi-node GB10 training throughput is not yet measured.

## Scaling to one billion atoms (with forces)

| GPU                   | Atoms per GPU        | GPUs, no halo | Halo factor | GPUs with halo          |
|-----------------------|----------------------|---------------|-------------|-------------------------|
| GB10                  | 54,872 (measured)    | 18,224        | 2.7x        | 48,702                  |
| H200 NVL              | 74,088 (measured)    | 13,497        | 2.5x        | 33,267                  |
| MI250X GCD (Frontier) | 33,917 (memory rule) | 29,484        | 3.1x        | 90,845 (= 11,356 nodes) |

The governing number

- With forces the force field needs 1.95 MB of GPU memory per atom on every card measured, so  $10^9$  atoms is  $\sim 2.0$  PB before overheads. 'Memory rule' rows apply that figure, and the 96% of memory the H200 could use, to the card's memory; they are estimates, not measurements.
- ALIGNN is a message-passing network, so when the crystal is split across GPUs each GPU also holds a ghost shell as thick as the receptive field ( $\sim 20$  Å here:  $2+2$  layers  $\times 5$  Å). Around a  $\sim 70$ k-atom Si block that multiplies the atoms each GPU holds by the halo factor. As-is, the GCD count exceeds Frontier (9,408 nodes).

Levers, largest first

- Memory per atom: the energy-only model uses 0.25 MB/atom (hidden 64, 12 neighbours); the 52-neighbour line graph dominates. A leaner force field, half precision and dropping stress could cut 1.95 MB several-fold (unmeasured, and accuracy must be re-checked).
- Fewer message-passing layers shrink the halo: 20 Å  $\rightarrow$  10 Å takes the factor from  $\sim 2.5$ x to  $\sim 1.6$ x.
- Memory per GPU matters more than speed: the job is memory-bound.

Software

- A run this size needs LAMMPS domain decomposition with ghost-atom exchange. ALIGNN has a LAMMPS interface (pair\_alignn, TorchScript); multi-layer ghost handling and an AMD build are not yet verified.
- The path is a ladder: 1 GPU  $\rightarrow$  1 node  $\rightarrow$  64 nodes  $\rightarrow$  thousands, fixing scaling at each step.

Time, if it fits

- At  $\sim 2.5$  s per step per GPU (H200 at its largest cell), 1 ps at 1 fs steps is  $\sim 42$  min of wall time plus communication; at thousands of nodes, every hour costs thousands of node-hours.

First steps (cheap, one GB10): memory per atom with forces for a leaner model (fewer neighbours, fewer layers, half precision), and pair\_alignn on 2 GPUs with a real ghost region.

## ALIGNN-FF results, energy only, B200 (1/2)

| Supercell | Atoms   | Neighbor list (s) | Graph (s) | Inference (s) | Total (s) | Model output (eV/atom) |
|-----------|---------|-------------------|-----------|---------------|-----------|------------------------|
| 1×1×1     | 4       | 0.0844            | 0.1568    | 3.2238        | 3.4650    | 0.604015               |
| 2×2×2     | 32      | 0.0015            | 0.0022    | 0.0073        | 0.0111    | 0.604014               |
| 3×3×3     | 108     | 0.0018            | 0.0026    | 0.0089        | 0.0132    | 0.604014               |
| 4×4×4     | 256     | 0.0087            | 0.0026    | 0.0100        | 0.0212    | 0.604014               |
| 5×5×5     | 500     | 0.0035            | 0.0029    | 0.0083        | 0.0147    | 0.604013               |
| 6×6×6     | 864     | 0.0047            | 0.0031    | 0.0086        | 0.0165    | 0.604013               |
| 7×7×7     | 1,372   | 0.0063            | 0.0035    | 0.0096        | 0.0194    | 0.604013               |
| 8×8×8     | 2,048   | 0.0085            | 0.0040    | 0.0109        | 0.0234    | 0.604013               |
| 9×9×9     | 2,916   | 0.0115            | 0.0046    | 0.0118        | 0.0279    | 0.604013               |
| 10×10×10  | 4,000   | 0.0158            | 0.0055    | 0.0136        | 0.0348    | 0.604013               |
| 11×11×11  | 5,324   | 0.0433            | 0.0181    | 0.0231        | 0.0845    | 0.604013               |
| 12×12×12  | 6,912   | 0.0680            | 0.0151    | 0.0304        | 0.1135    | 0.604013               |
| 13×13×13  | 8,788   | 0.0733            | 0.0248    | 0.0476        | 0.1457    | 0.604013               |
| 14×14×14  | 10,976  | 0.0881            | 0.0216    | 0.0542        | 0.1639    | 0.604013               |
| 15×15×15  | 13,500  | 0.1020            | 0.0252    | 0.0498        | 0.1771    | 0.604013               |
| 16×16×16  | 16,384  | 0.1218            | 0.0323    | 0.0682        | 0.2223    | 0.604013               |
| 17×17×17  | 19,652  | 0.1479            | 0.0399    | 0.0669        | 0.2547    | 0.604013               |
| 18×18×18  | 23,328  | 0.1598            | 0.0403    | 0.0786        | 0.2787    | 0.604013               |
| 19×19×19  | 27,436  | 0.1912            | 0.0387    | 0.0928        | 0.3226    | 0.604013               |
| 20×20×20  | 32,000  | 0.2106            | 0.0414    | 0.0985        | 0.3506    | 0.604013               |
| 21×21×21  | 37,044  | 0.2482            | 0.0462    | 0.1152        | 0.4096    | 0.604013               |
| 22×22×22  | 42,592  | 0.2590            | 0.0495    | 0.1283        | 0.4368    | 0.604013               |
| 23×23×23  | 48,668  | 0.2798            | 0.0548    | 0.1499        | 0.4845    | 0.604013               |
| 24×24×24  | 55,296  | 0.2096            | 0.0429    | 0.1373        | 0.3897    | 0.604013               |
| 25×25×25  | 62,500  | 0.2265            | 0.0481    | 0.1548        | 0.4294    | 0.604013               |
| 26×26×26  | 70,304  | 0.2547            | 0.0537    | 0.1738        | 0.4822    | 0.604015               |
| 27×27×27  | 78,732  | 0.2841            | 0.0601    | 0.1935        | 0.5377    | 0.604013               |
| 28×28×28  | 87,808  | 0.3112            | 0.0651    | 0.2156        | 0.5919    | 0.604013               |
| 29×29×29  | 97,556  | 0.3462            | 0.0750    | 0.2390        | 0.6602    | 0.604013               |
| 30×30×30  | 108,000 | 0.3773            | 0.0811    | 0.2643        | 0.7227    | 0.604013               |

ALIGNN-FF v6 (scale5b\_v6.py), one B200, float32. Times are wall time for one energy evaluation.  
The first row includes one-time GPU warm-up. Red rows (N > 442,368): the model output is wrong because of the float32 drift; the float64 reference is 0.604015 eV/atom.

## ALIGNN-FF results, energy only, B200 (2/2)

| Supercell | Atoms   | Neighbor list (s) | Graph (s) | Inference (s) | Total (s) | Model output (eV/atom) |
|-----------|---------|-------------------|-----------|---------------|-----------|------------------------|
| 31×31×31  | 119,164 | 0.4516            | 0.0884    | 0.2913        | 0.8313    | 0.604013               |
| 32×32×32  | 131,072 | 0.4864            | 0.0953    | 0.3193        | 0.9011    | 0.604013               |
| 33×33×33  | 143,748 | 0.5267            | 0.1030    | 0.3502        | 0.9799    | 0.604013               |
| 34×34×34  | 157,216 | 0.5839            | 0.1139    | 0.3820        | 1.0798    | 0.604013               |
| 35×35×35  | 171,500 | 0.6101            | 0.1259    | 0.4164        | 1.1525    | 0.604013               |
| 36×36×36  | 186,624 | 0.6554            | 0.1350    | 0.4529        | 1.2434    | 0.604013               |
| 37×37×37  | 202,612 | 0.7091            | 0.1466    | 0.4912        | 1.3469    | 0.604013               |
| 38×38×38  | 219,488 | 0.7572            | 0.1677    | 0.5319        | 1.4568    | 0.604013               |
| 39×39×39  | 237,276 | 0.8124            | 0.1740    | 0.5753        | 1.5617    | 0.604013               |
| 40×40×40  | 256,000 | 0.8806            | 0.1872    | 0.6202        | 1.6880    | 0.604013               |
| 41×41×41  | 275,684 | 1.0001            | 0.1986    | 0.6675        | 1.8662    | 0.604013               |
| 42×42×42  | 296,352 | 1.0695            | 0.2132    | 0.7176        | 2.0003    | 0.604013               |
| 43×43×43  | 318,028 | 1.1327            | 0.2275    | 0.7691        | 2.1293    | 0.604015               |
| 44×44×44  | 340,736 | 1.1961            | 0.2483    | 0.8239        | 2.2683    | 0.604013               |
| 45×45×45  | 364,500 | 1.2776            | 0.2635    | 0.8810        | 2.4221    | 0.604013               |
| 46×46×46  | 389,344 | 1.3576            | 0.2852    | 0.9407        | 2.5835    | 0.604015               |
| 47×47×47  | 415,292 | 1.4514            | 0.3117    | 1.0032        | 2.7663    | 0.604015               |
| 48×48×48  | 442,368 | 1.5614            | 0.3171    | 1.0683        | 2.9467    | 0.604013               |
| 49×49×49  | 470,596 | 1.6539            | 0.3416    | 1.1366        | 3.1321    | 0.606979               |
| 50×50×50  | 500,000 | 1.7436            | 0.3695    | 1.2075        | 3.3206    | 0.639071               |
| 51×51×51  | 530,604 | 1.9523            | 0.3813    | 1.2820        | 3.6156    | 0.672090               |
| 52×52×52  | 562,432 | 2.0831            | 0.4097    | 1.3590        | 3.8518    | 0.702621               |
| 53×53×53  | 595,508 | 2.1964            | 0.4336    | 1.4393        | 4.0692    | 0.730894               |
| 54×54×54  | 629,856 | 2.3269            | 0.4530    | 1.5224        | 4.3024    | 0.757119               |
| 55×55×55  | 665,500 | 2.4602            | 0.4827    | 1.6091        | 4.5520    | 0.781473               |
| 56×56×56  | 702,464 | 2.5851            | 0.5138    | 1.6982        | 4.7971    | 0.804124               |
| 57×57×57  | 740,772 | 2.7414            | 0.5370    | 1.7917        | 5.0702    | 0.825215               |
| 58×58×58  | 780,448 | 2.9055            | 0.5709    | 1.9083        | 5.3847    | 0.844881               |

ALIGNN-FF v6 (scale5b\_v6.py), one B200, float32. Times are wall time for one energy evaluation.  
The first row includes one-time GPU warm-up. Red rows (N > 442,368): the model output is wrong because of the float32 drift; the float64 reference is 0.604015 eV/atom.

ALIGNN-FF results, energy only, GB10 (1/2)

| Supercell | Atoms   | Graph (s) | Inference (s) | Total (s) | Peak mem (GB) | Model output (eV/atom) |
|-----------|---------|-----------|---------------|-----------|---------------|------------------------|
| 1×1×1     | 4       | 1.3637    | 0.0701        | 1.4338    | 0.0           | 0.604014               |
| 2×2×2     | 32      | 0.0017    | 0.0025        | 0.0042    | 0.0           | 0.604015               |
| 3×3×3     | 108     | 0.0145    | 0.0037        | 0.0182    | 0.1           | 0.604015               |
| 4×4×4     | 256     | 0.0050    | 0.0066        | 0.0115    | 0.1           | 0.604020               |
| 5×5×5     | 500     | 0.0053    | 0.0120        | 0.0173    | 0.2           | 0.604017               |
| 6×6×6     | 864     | 0.0068    | 0.0189        | 0.0257    | 0.3           | 0.604021               |
| 7×7×7     | 1,372   | 0.0087    | 0.0287        | 0.0374    | 0.4           | 0.604014               |
| 8×8×8     | 2,048   | 0.0122    | 0.0438        | 0.0560    | 0.6           | 0.604022               |
| 9×9×9     | 2,916   | 0.0148    | 0.0639        | 0.0787    | 0.8           | 0.604024               |
| 10×10×10  | 4,000   | 0.0195    | 0.0865        | 0.1060    | 1.0           | 0.604012               |
| 11×11×11  | 5,324   | 0.0279    | 0.1159        | 0.1437    | 1.4           | 0.604015               |
| 12×12×12  | 6,912   | 0.0326    | 0.1538        | 0.1864    | 1.8           | 0.604044               |
| 13×13×13  | 8,788   | 0.0367    | 0.1924        | 0.2291    | 2.3           | 0.604008               |
| 14×14×14  | 10,976  | 0.0458    | 0.2384        | 0.2841    | 2.8           | 0.603933               |
| 15×15×15  | 13,500  | 0.0608    | 0.2931        | 0.3539    | 3.5           | 0.603850               |
| 16×16×16  | 16,384  | 0.0770    | 0.3540        | 0.4310    | 4.2           | 0.603873               |
| 17×17×17  | 19,652  | 0.0910    | 0.4241        | 0.5151    | 5.0           | 0.603919               |
| 18×18×18  | 23,328  | 0.1104    | 0.5035        | 0.6139    | 5.9           | 0.603956               |
| 19×19×19  | 27,436  | 0.1265    | 0.5906        | 0.7171    | 7.0           | 0.603956               |
| 20×20×20  | 32,000  | 0.1427    | 0.6841        | 0.8268    | 8.1           | 0.603860               |
| 21×21×21  | 37,044  | 0.1612    | 0.7881        | 0.9493    | 9.4           | 0.603743               |
| 22×22×22  | 42,592  | 0.1921    | 0.9065        | 1.0986    | 10.8          | 0.603745               |
| 23×23×23  | 48,668  | 0.2169    | 1.0358        | 1.2527    | 12.4          | 0.603927               |
| 24×24×24  | 55,296  | 0.2349    | 1.1735        | 1.4083    | 14.0          | 0.604198               |
| 25×25×25  | 62,500  | 0.2678    | 1.3306        | 1.5985    | 15.9          | 0.604642               |
| 26×26×26  | 70,304  | 0.3060    | 1.4957        | 1.8016    | 17.8          | 0.605064               |
| 27×27×27  | 78,732  | 0.3374    | 1.6727        | 2.0101    | 20.0          | 0.605457               |
| 28×28×28  | 87,808  | 0.3540    | 1.8783        | 2.2323    | 22.3          | 0.605647               |
| 29×29×29  | 97,556  | 0.4018    | 2.0760        | 2.4778    | 24.8          | 0.605734               |
| 30×30×30  | 108,000 | 0.4980    | 2.2980        | 2.7959    | 27.4          | 0.605611               |

B200 settings rerun on one NVIDIA GB10: Cu FCC, 5 Å cutoff, 12 neighbours, energy only, float32, checkpoint v12.2.2024\_dft\_3d\_307k, loaded into the pure-PyTorch ALIGNN (reproduces the B200 DGL energies to 4e-6 eV/atom). One call per size; the first row includes GPU warm-up. Largest size shown is the largest that completed; the next (49×49×49 = 470,596 atoms) did not fit. Red rows (N > 442,368): float32 energy drift, as on the B200. The port has no float64 readout, so smaller cells also drift slightly (to ~0.600). Timings are unaffected. atomgptlab job 13484, 2026-09-28.

## ALIGNN-FF results, energy only, GB10 (2/2)

| Supercell | Atoms   | Graph (s) | Inference (s) | Total (s) | Peak mem (GB) | Model output (eV/atom) |
|-----------|---------|-----------|---------------|-----------|---------------|------------------------|
| 31×31×31  | 119,164 | 0.5406    | 2.5462        | 3.0868    | 30.2          | 0.604993               |
| 32×32×32  | 131,072 | 0.5858    | 2.7901        | 3.3759    | 33.2          | 0.604449               |
| 33×33×33  | 143,748 | 0.6294    | 3.1341        | 3.7635    | 36.5          | 0.603988               |
| 34×34×34  | 157,216 | 0.6864    | 3.3575        | 4.0439    | 39.9          | 0.603673               |
| 35×35×35  | 171,500 | 0.7332    | 3.6670        | 4.4002    | 43.5          | 0.602742               |
| 36×36×36  | 186,624 | 0.8208    | 3.9880        | 4.8089    | 47.3          | 0.602057               |
| 37×37×37  | 202,612 | 0.8541    | 4.4270        | 5.2811    | 51.4          | 0.601320               |
| 38×38×38  | 219,488 | 0.9349    | 4.6847        | 5.6196    | 55.6          | 0.601225               |
| 39×39×39  | 237,276 | 1.0287    | 5.1840        | 6.2126    | 60.1          | 0.600929               |
| 40×40×40  | 256,000 | 1.1166    | 5.4838        | 6.6004    | 64.9          | 0.600853               |
| 41×41×41  | 275,684 | 1.1886    | 5.8977        | 7.0864    | 69.9          | 0.600795               |
| 42×42×42  | 296,352 | 1.2488    | 6.3326        | 7.5814    | 75.1          | 0.600620               |
| 43×43×43  | 318,028 | 1.3689    | 6.8675        | 8.2364    | 80.6          | 0.600555               |
| 44×44×44  | 340,736 | 1.4533    | 7.3131        | 8.7664    | 86.4          | 0.600509               |
| 45×45×45  | 364,500 | 1.4925    | 8.0306        | 9.5232    | 92.4          | 0.600274               |
| 46×46×46  | 389,344 | 1.7065    | 8.4010        | 10.1075   | 98.7          | 0.600081               |
| 47×47×47  | 415,292 | 1.8319    | 8.9564        | 10.7883   | 105.2         | 0.599898               |
| 48×48×48  | 442,368 | 1.9177    | 17.4482       | 19.3659   | 112.1         | 0.600148               |

B200 settings rerun on one NVIDIA GB10: Cu FCC, 5 Å cutoff, 12 neighbours, energy only, float32, checkpoint v12.2.2024\_dft\_3d\_307k, loaded into the pure-PyTorch ALIGNN (reproduces the B200 DGL energies to 4e-6 eV/atom). One call per size; the first row includes GPU warm-up. Largest size shown is the largest that completed; the next (49×49×49 = 470,596 atoms) did not fit. Red rows (N > 442,368): float32 energy drift, as on the B200. The port has no float64 readout, so smaller cells also drift slightly (to ~0.600). Timings are unaffected. atomgptlab job 13484, 2026-09-28.

## ALIGNN-FF results with forces, GB10

| Supercell | Atoms  | Time (s) | Peak mem (GB) | MB/atom | Energy (eV/atom) | Max  F  (eV/A) |
|-----------|--------|----------|---------------|---------|------------------|----------------|
| 2×2×2     | 64     | 0.043    | 0.2           | 3.03    | -8.7476          | 2.0e-05        |
| 4×4×4     | 512    | 0.114    | 1.1           | 2.09    | -8.7476          | 3.5e-05        |
| 6×6×6     | 1,728  | 0.353    | 3.4           | 1.99    | -8.7476          | 3.8e-05        |
| 8×8×8     | 4,096  | 0.635    | 8.1           | 1.97    | -8.7475          | 3.8e-05        |
| 10×10×10  | 8,000  | 1.174    | 15.7          | 1.96    | -8.7479          | 3.9e-05        |
| 12×12×12  | 13,824 | 1.959    | 27.0          | 1.96    | -8.7479          | 3.9e-05        |
| 14×14×14  | 21,952 | 3.116    | 42.9          | 1.95    | -8.7476          | 5.6e-05        |
| 15×15×15  | 27,000 | 3.742    | 52.7          | 1.95    | -8.7470          | 5.7e-05        |
| 16×16×16  | 32,768 | 4.576    | 64.0          | 1.95    | -8.7464          | 5.8e-05        |
| 17×17×17  | 39,304 | 5.504    | 76.8          | 1.95    | -8.7467          | 5.7e-05        |
| 18×18×18  | 46,656 | 6.454    | 91.1          | 1.95    | -8.7470          | 5.7e-05        |
| 19×19×19  | 54,872 | 7.592    | 107.1         | 1.95    | -8.7475          | 5.7e-05        |
| 20×20×20  | 64,000 | OOM      | -             | -       | -                | -              |

Energy + forces + stress single point on one NVIDIA GB10: Si diamond (8-atom conventional cell,  $i \times i \times i$ ), default matpes\_r2scan checkpoint (hidden 128, 2+2 layers, smooth 5 Å cutoff, 52 neighbours), ASE calculator, float32. Time = median of 3 calls. Peak mem = peak allocated GPU memory. Max |F| should be  $\sim 0$  for the perfect crystal (checks the forces). Red row: the first size that ran out of GPU memory. atomgptlab job 13474, 2026-09-27.

ALIGNN-FF results, energy only, H200 (1/2)

| Supercell | Atoms   | Graph (s) | Inference (s) | Total (s) | Peak mem (GB) | Model output (eV/atom) |
|-----------|---------|-----------|---------------|-----------|---------------|------------------------|
| 1×1×1     | 4       | 0.5613    | 0.0668        | 0.6281    | 0.0           | 0.604015               |
| 2×2×2     | 32      | 0.0024    | 0.0023        | 0.0046    | 0.0           | 0.604017               |
| 3×3×3     | 108     | 0.0074    | 0.0025        | 0.0099    | 0.1           | 0.604017               |
| 4×4×4     | 256     | 0.0051    | 0.0025        | 0.0076    | 0.1           | 0.604020               |
| 5×5×5     | 500     | 0.0180    | 0.0032        | 0.0211    | 0.2           | 0.604019               |
| 6×6×6     | 864     | 0.0298    | 0.0035        | 0.0333    | 0.3           | 0.604021               |
| 7×7×7     | 1,372   | 0.0316    | 0.0049        | 0.0365    | 0.4           | 0.604014               |
| 8×8×8     | 2,048   | 0.0382    | 0.0067        | 0.0449    | 0.6           | 0.604023               |
| 9×9×9     | 2,916   | 0.0411    | 0.0093        | 0.0504    | 0.8           | 0.604024               |
| 10×10×10  | 4,000   | 0.0441    | 0.0121        | 0.0562    | 1.0           | 0.604011               |
| 11×11×11  | 5,324   | 0.0619    | 0.0157        | 0.0776    | 1.4           | 0.604010               |
| 12×12×12  | 6,912   | 0.0645    | 0.0203        | 0.0848    | 1.8           | 0.604036               |
| 13×13×13  | 8,788   | 0.0721    | 0.0254        | 0.0976    | 2.3           | 0.604000               |
| 14×14×14  | 10,976  | 0.0987    | 0.0313        | 0.1301    | 2.8           | 0.603929               |
| 15×15×15  | 13,500  | 0.1098    | 0.0382        | 0.1480    | 3.5           | 0.603846               |
| 16×16×16  | 16,384  | 0.1138    | 0.0460        | 0.1598    | 4.2           | 0.603869               |
| 17×17×17  | 19,652  | 0.1256    | 0.0561        | 0.1818    | 5.0           | 0.603917               |
| 18×18×18  | 23,328  | 0.1608    | 0.0648        | 0.2257    | 5.9           | 0.603956               |
| 19×19×19  | 27,436  | 0.1737    | 0.0762        | 0.2499    | 7.0           | 0.603955               |
| 20×20×20  | 32,000  | 0.1851    | 0.0884        | 0.2735    | 8.1           | 0.603858               |
| 21×21×21  | 37,044  | 0.2089    | 0.1022        | 0.3110    | 9.4           | 0.603743               |
| 22×22×22  | 42,592  | 0.2497    | 0.1174        | 0.3671    | 10.8          | 0.603744               |
| 23×23×23  | 48,668  | 0.2379    | 0.1339        | 0.3718    | 12.4          | 0.603927               |
| 24×24×24  | 55,296  | 0.2551    | 0.1519        | 0.4070    | 14.0          | 0.604197               |
| 25×25×25  | 62,500  | 0.2764    | 0.1720        | 0.4484    | 15.9          | 0.604641               |
| 26×26×26  | 70,304  | 0.3377    | 0.1929        | 0.5306    | 17.8          | 0.605063               |
| 27×27×27  | 78,732  | 0.3614    | 0.2159        | 0.5773    | 20.0          | 0.605456               |
| 28×28×28  | 87,808  | 0.3882    | 0.2408        | 0.6291    | 22.3          | 0.605646               |
| 29×29×29  | 97,556  | 0.5792    | 0.2674        | 0.8466    | 24.8          | 0.605734               |
| 30×30×30  | 108,000 | 0.6158    | 0.2960        | 0.9118    | 27.4          | 0.605611               |

B200 settings rerun on one NVIDIA H200 NVL: Cu FCC, 5 Å cutoff, 12 neighbours, energy only, float32, checkpoint v12.2.2024\_dft\_3d\_307k, loaded into the pure-PyTorch ALIGNN (reproduces the B200 DGL energies to 4e-6 eV/atom). One call per size; the first row includes GPU warm-up. Largest size shown is the largest that completed; the next (53×53×53 = 595,508 atoms) did not fit. Red rows (N > 442,368): float32 energy drift, as on the B200. The port has no float64 readout, so smaller cells also drift slightly (to ~0.600). Timings are unaffected. Skipjack job 920089, 2026-09-28.

## ALIGNN-FF results, energy only, H200 (2/2)

| Supercell | Atoms   | Graph (s) | Inference (s) | Total (s) | Peak mem (GB) | Model output (eV/atom) |
|-----------|---------|-----------|---------------|-----------|---------------|------------------------|
| 31×31×31  | 119,164 | 0.6895    | 0.3260        | 1.0156    | 30.2          | 0.604993               |
| 32×32×32  | 131,072 | 0.7279    | 0.3587        | 1.0866    | 33.2          | 0.604449               |
| 33×33×33  | 143,748 | 0.8287    | 0.3946        | 1.2233    | 36.5          | 0.603987               |
| 34×34×34  | 157,216 | 0.8357    | 0.4293        | 1.2650    | 39.9          | 0.603673               |
| 35×35×35  | 171,500 | 0.8934    | 0.4685        | 1.3619    | 43.5          | 0.602742               |
| 36×36×36  | 186,624 | 1.0349    | 0.5095        | 1.5444    | 47.3          | 0.602057               |
| 37×37×37  | 202,612 | 1.1004    | 0.5565        | 1.6569    | 51.4          | 0.601320               |
| 38×38×38  | 219,488 | 1.2563    | 0.5999        | 1.8562    | 55.6          | 0.601225               |
| 39×39×39  | 237,276 | 1.3330    | 0.6514        | 1.9845    | 60.1          | 0.600928               |
| 40×40×40  | 256,000 | 1.5010    | 0.7002        | 2.2011    | 64.9          | 0.600853               |
| 41×41×41  | 275,684 | 1.5399    | 0.7535        | 2.2934    | 69.9          | 0.600795               |
| 42×42×42  | 296,352 | 1.5578    | 0.8105        | 2.3683    | 75.1          | 0.600620               |
| 43×43×43  | 318,028 | 1.7216    | 0.8683        | 2.5899    | 80.6          | 0.600554               |
| 44×44×44  | 340,736 | 1.8298    | 0.9311        | 2.7609    | 86.4          | 0.600510               |
| 45×45×45  | 364,500 | 2.0288    | 1.0003        | 3.0292    | 92.4          | 0.600274               |
| 46×46×46  | 389,344 | 2.1526    | 1.0680        | 3.2206    | 98.7          | 0.600081               |
| 47×47×47  | 415,292 | 2.3286    | 1.1412        | 3.4699    | 105.3         | 0.599898               |
| 48×48×48  | 442,368 | 2.4385    | 1.2090        | 3.6476    | 112.1         | 0.600148               |
| 49×49×49  | 470,596 | 2.4171    | 1.4137        | 3.8308    | 119.3         | 0.602111               |
| 50×50×50  | 500,000 | 2.5177    | 1.5239        | 4.0417    | 126.7         | 0.619691               |
| 51×51×51  | 530,604 | 2.7645    | 1.6988        | 4.4633    | 134.5         | 0.639844               |
| 52×52×52  | 562,432 | 2.9893    | 1.9614        | 4.9507    | 142.5         | 0.658485               |

B200 settings rerun on one NVIDIA H200 NVL: Cu FCC, 5 Å cutoff, 12 neighbours, energy only, float32, checkpoint v12.2.2024\_dft\_3d\_307k, loaded into the pure-PyTorch ALIGNN (reproduces the B200 DGL energies to 4e-6 eV/atom). One call per size; the first row includes GPU warm-up. Largest size shown is the largest that completed; the next (53×53×53 = 595,508 atoms) did not fit. Red rows (N > 442,368): float32 energy drift, as on the B200. The port has no float64 readout, so smaller cells also drift slightly (to ~0.600). Timings are unaffected. Skipjack job 920089, 2026-09-28.

## ALIGNN-FF results with forces, H200

| Supercell | Atoms  | Time (s) | Peak mem (GB) | MB/atom | Energy (eV/atom) | Max  F  (eV/A) |
|-----------|--------|----------|---------------|---------|------------------|----------------|
| 2×2×2     | 64     | 0.020    | 0.2           | 3.05    | -8.7476          | 2.2e-05        |
| 4×4×4     | 512    | 0.032    | 1.1           | 2.09    | -8.7476          | 3.6e-05        |
| 6×6×6     | 1,728  | 0.075    | 3.4           | 1.99    | -8.7476          | 3.7e-05        |
| 8×8×8     | 4,096  | 0.159    | 8.1           | 1.97    | -8.7475          | 3.9e-05        |
| 10×10×10  | 8,000  | 0.278    | 15.7          | 1.96    | -8.7479          | 3.8e-05        |
| 12×12×12  | 13,824 | 0.470    | 27.0          | 1.96    | -8.7479          | 4.0e-05        |
| 14×14×14  | 21,952 | 0.724    | 42.9          | 1.95    | -8.7476          | 5.6e-05        |
| 15×15×15  | 27,000 | 0.905    | 52.7          | 1.95    | -8.7470          | 5.8e-05        |
| 16×16×16  | 32,768 | 1.175    | 64.0          | 1.95    | -8.7464          | 5.6e-05        |
| 17×17×17  | 39,304 | 1.339    | 76.8          | 1.95    | -8.7467          | 5.7e-05        |
| 18×18×18  | 46,656 | 1.578    | 91.1          | 1.95    | -8.7470          | 5.7e-05        |
| 19×19×19  | 54,872 | 1.887    | 107.1         | 1.95    | -8.7475          | 5.6e-05        |
| 20×20×20  | 64,000 | 2.184    | 124.9         | 1.95    | -8.7482          | 5.8e-05        |
| 21×21×21  | 74,088 | 2.534    | 144.6         | 1.95    | -8.7482          | 5.8e-05        |
| 22×22×22  | 85,184 | OOM      | -             | -       | -                | -              |

Energy + forces + stress single point on one NVIDIA H200 NVL: Si diamond (8-atom conventional cell,  $i \times i \times i$ ), default matpes\_r2scan checkpoint (hidden 128, 2+2 layers, smooth 5 Å cutoff, 52 neighbours), ASE calculator, float32. Time = median of 3 calls. Peak mem = peak allocated GPU memory. Max |F| should be  $\sim 0$  for the perfect crystal (checks the forces). Red row: the first size that ran out of GPU memory. Skipjack job 913733, 2026-09-27.

# VASP results

| Supercell | Atoms | GPUs | k-points | Bands  | SCF cycles | Elapsed (s) | s / SCF | E (eV/atom) | Status        |
|-----------|-------|------|----------|--------|------------|-------------|---------|-------------|---------------|
| 3×3×3     | 54    | 1    | 3        | 136    | 20/20      | 15.3        | 0.76    | -5.4105     | converged     |
| 3×3×3     | 54    | 2    | 3        | 136    | 20/20      | 17.8        | 0.89    | -5.4105     | converged     |
| 3×3×3     | 54    | 4    | 3        | 136    | 20/20      | 18.3        | 0.91    | -5.4105     | converged     |
| 3×3×3     | 54    | 8    | 3        | 136    | 20/20      | 27.3        | 1.36    | -5.4105     | converged     |
| 4×4×4     | 128   | 1    | 3        | 321    | 20/20      | 39.6        | 1.98    | -5.4211     | converged     |
| 4×4×4     | 128   | 2    | 3        | 321    | 20/20      | 38.3        | 1.92    | -5.4211     | converged     |
| 4×4×4     | 128   | 4    | 3        | 321    | 20/20      | 36.5        | 1.82    | -5.4211     | converged     |
| 4×4×4     | 128   | 8    | 3        | 321    | 20/20      | 42.5        | 2.13    | -5.4211     | converged     |
| 5×5×5     | 250   | 1    | 1        | 626    | 20/20      | 57.6        | 2.88    | -5.3903     | converged     |
| 5×5×5     | 250   | 2    | 1        | 626    | 20/20      | 62.4        | 3.12    | -5.3903     | converged     |
| 5×5×5     | 250   | 4    | 1        | 628    | 20/20      | 75.3        | 3.76    | -5.3903     | converged     |
| 5×5×5     | 250   | 8    | 1        | 632    | 20/20      | 108.1       | 5.41    | -5.3903     | converged     |
| 6×6×6     | 432   | 1    | 1        | 1,081  | 20/20      | 113.7       | 5.68    | -5.4111     | converged     |
| 6×6×6     | 432   | 2    | 1        | 1,082  | 20/20      | 96.2        | 4.81    | -5.4111     | converged     |
| 6×6×6     | 432   | 4    | 1        | 1,084  | 20/20      | 89.3        | 4.46    | -5.4111     | converged     |
| 6×6×6     | 432   | 8    | 1        | 1,088  | 20/20      | 123.6       | 6.18    | -5.4111     | converged     |
| 10×10×10  | 2,000 | 4    | 1        | 5,004  | 5/5        | 518.4       | 103.69  | -5.5124     | not converged |
| 12×12×12  | 3,456 | 8    | 1        | 8,648  | 2/2        | 678.6       | 339.29  | -3.1397     | not converged |
| 14×14×14  | 5,488 | 8    | 1        | 13,728 | 2/2        | 1561.4      | 780.69  | -3.1433     | not converged |
| 15×15×15  | 6,750 | 8    | 1        | 16,880 | 0/2        | -           | -       | -           | crashed       |
| 16×16×16  | 8,192 | 8    | 1        | 20,488 | 0/2        | -           | -       | -           | crashed       |

VASP GPU build, single-point SCF. 3×3×3 to 6×6×6: NELM=20, EDIFF=1E-6. 10×10×10: NELM=5, ALGO=Fast. 12×12×12 and larger: NELM=2, ALGO=Fast. SCF cycles = completed / maximum allowed (NELM). E = energy of the last SCF cycle / atom count. Converged = last energy change below the run's EDIFF. Amber rows were stopped early for timing; their E is an unconverged intermediate value, not a result. Red rows crashed before the first SCF cycle (likely out of memory).